Vertical Flight Infrastructure Data Quality Shortcomings

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FORUM80
THE FUTURE OF VERTICAL FLIGHT
MAY 7-9, 2024 MONTRÉAL, QUÉBEC, CANADA

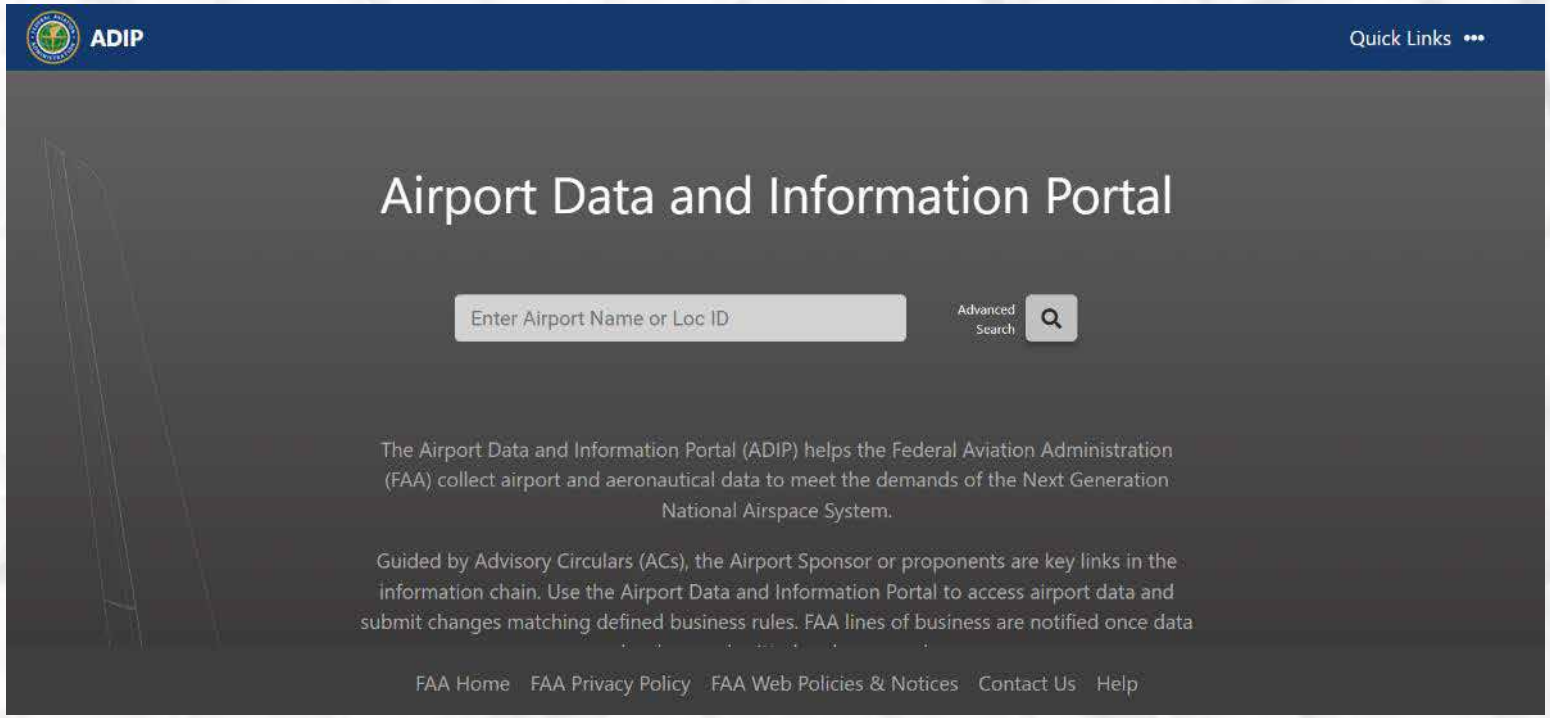
Goal

Conduct a review of vertical flight infrastructure data contained within the Federal Aviation Administration (FAA) Airport Master Record (AMR) for the purpose of evaluating the Data Collection process and to what extent it is being successfully accomplished in regard to Accuracy, Resolution, Integrity, Traceability, Timeliness, Completeness, and Format.

Database Utilized

Federal Aviation Administration (FAA)

- Airport Data Information Portal (ADIP) https://adip.faa.gov



Oversight and Responsibility

- 49 U.S.C. 329(b) empowers and directs the Secretary of Transportation to collect and disseminate information on civil aeronautics.
- Aeronautical information is required by the FAA in order to carry out a multitude of agency missions and functions.
- However, for private use facilities this responsibility resides with the owner of the facility.

Accuracy Implications

Users of the FAA data include but are not limited to Jeppesen, ForeFlight, GPS Manufactures, Aircraft Owners, Pilots, Aviation Associations, State Agencies, State Airport Directories, Aeronautical Publications, System Planners, and other government agencies such as the National Oceanic & Atmospheric Administration and the National Geospatial-Intelligence Agency.

Garmin marketing material for a portable aviation GPS, “aera 796”.

“…while a private airport and heliport database greatly expands your emergency “nearest” options.”

ASIAS (Aviation Safety Information Analysis and Sharing)

A collaborative government and industry initiative on data sharing and analysis to proactively discover safety concerns before accidents or incidents occur, leading to timely mitigation and prevention

https://www.asias.faa.gov

ASIAS is Governed by Formal Principles

Data used solely for **advancement of safety**

Voluntary submission of safety-sensitive data

Operator/OEM/MRO **data are de-identified**

Transparency for how data are managed and utilized

Procedures & policies based on **collaborative governance**

Analyses **approved by** an **ASIAS Executive Board**

The Importance of Good Data for ASIAS

U.S. DOT/FAA Heliport Guide AC 150/5390-2D

Prior Permission Required (PPR) Heliport.

A heliport developed for exclusive use of the owner and persons authorized by the owner and about which the owner and operator ensure all authorized pilots are thoroughly knowledgeable. These features include but are not limited to:

- *approach/departure path characteristics*
- *preferred heading*
- *lighting*
- *obstacles in the area*
- *size and weight capacity of the facility*
- *heliport facility limitations*

U.S. Aviation Infrastructure Breakdown

Infrastructure By Type	Number
Airports	13,154
Balloonports	13
Gliderports	36
Heliports	6,185
Seaplane Bases	541
Ultralight Parks	112
Vertiports	0
As of March 2024* **Total U.S. Heliports	**20,032**

Ownership Findings

Heliport Ownership	Number
Public Owned	635
Private Owned	5,454
U.S. Coast Guard	4
U.S. Air Force	6
U.S. Navy	14
U.S. Army	72
As of March 2024 **Total U.S. Heliports**	**6,185**

*It is estimated there are 131 heliports at airports. Those numbers are not included here.

Use Case Findings

Heliport Use Case	Number
Public-Use*	55
Private-Use	6,130
*As of March 2024 *Medical-Use***	*2,849*

**Less than 1% of all Heliports in the U.S. are identified as Public-Use*

***Accounting for 46% of all heliports in the U.S., "Medical-Use" is a subcategory for which there are 3 public use and 2,846 private use heliports.*

***At least 332 healthcare facilities were not accounted for as Medical Use in the ADIP AMR Database.*

Hospitals

It has been found that the majority of hospitals in the U.S. have a heliport associated with it for transporting patients.

U.S. Hospitals	Number
Community Hospitals	5,129
Federal Government Hospitals	207
Nonfederal Psychiatric Hospitals	659
Other Hospitals	125
As of March 2024 **Total:**	**6,120**

Reference: American Hospital Association Website https://www.aha.org/statistics/fast-facts-us-hospitals

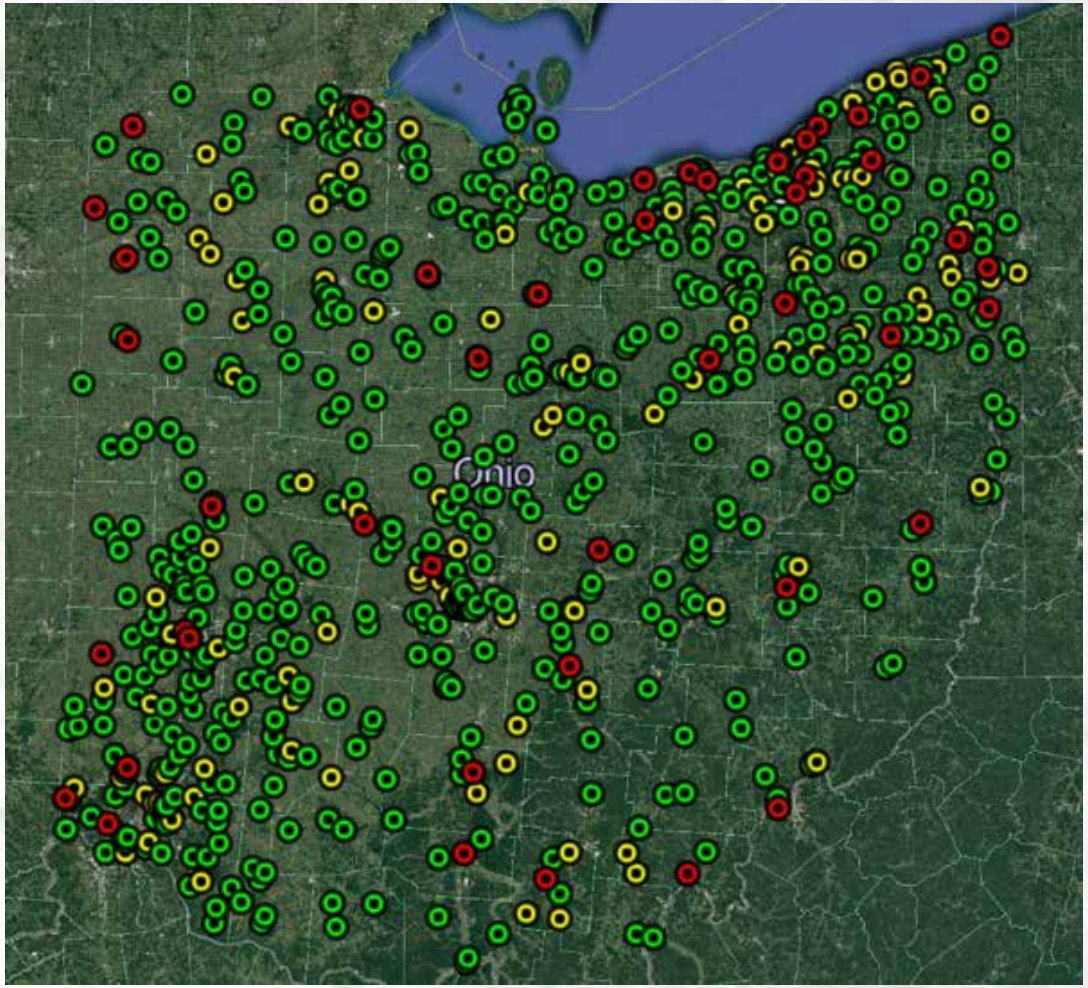
Missing Hospital Heliports

In 2019, NASA determined that there was an estimated underreporting of U.S. hospital heliport of upwards of **2,000** in the FAA's AMR database system.

NASA Aviation Safety Report, ASRS No. 1599969, Jan 31, 2019

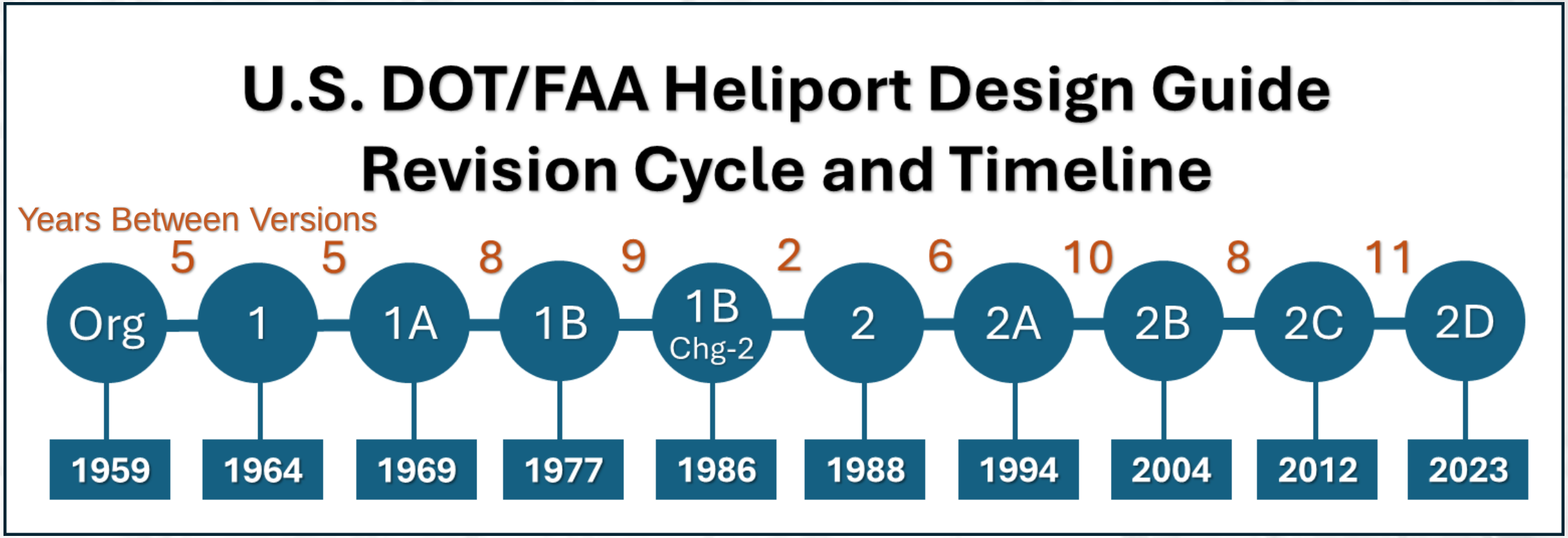
Type	No.
Airports	428
Heliports	231
Seaplane Bases	3
Gliderports	2
Balloonports	1
Total	665

Verified	537
Inaccurate or Missing	128
Unlisted/Unknown	44



Results of data 2018 accountability study conducted by HeliExperts International for the state of Ohio

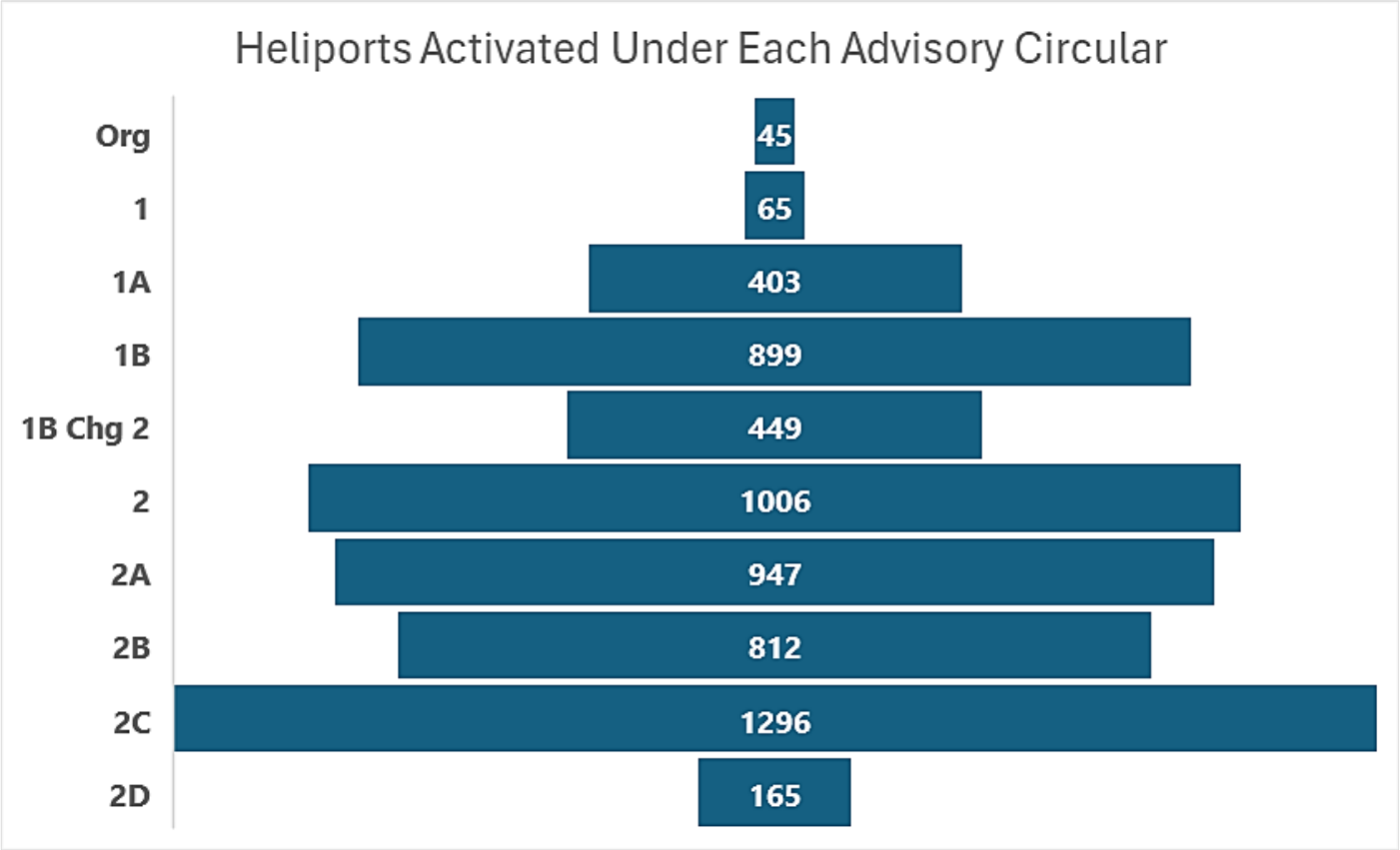
Evolution of the FAA Heliport Guide, AC 150/5390



Heliports Activated by Circular

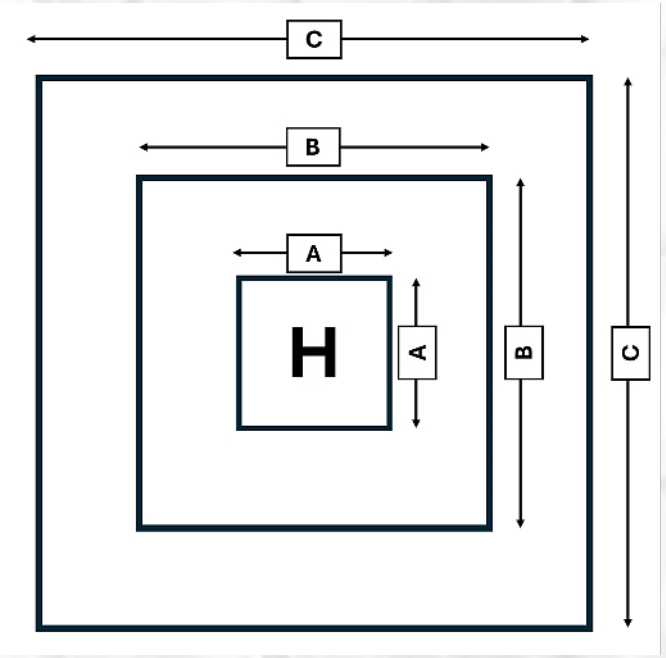
This is an approximation based on ADIP AMR Data. It is estimated that there is a potential 3-5% error given that many heliports are issued an airspace determination letter which is good for 18-months with the potential for at least one 6-month extension. Activation dates are generally issued after a heliport has been built but can be retroactive to the advisory circular it was submitted under.

There were 12 heliports on record as having existed prior to 1959, with the earliest one having an activation date of 1927. These were not included in this study.



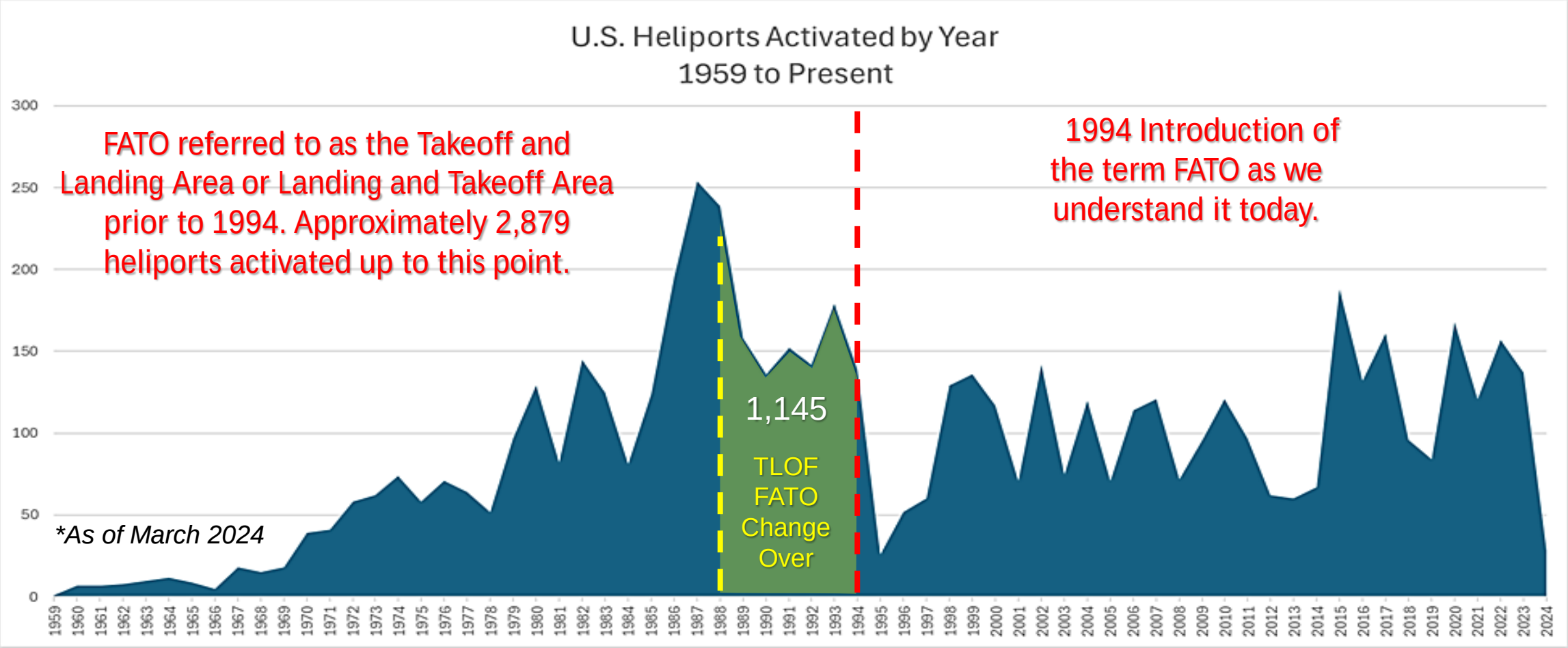
Heliport Terminology over the Years

AC	Published	A	B	C
2D	Jan 2023	TLOF	FATO	Safety Area
2C	Apr 2012	TLOF	FATO	Safety Area
2B	Sep 2004	TLOF	FATO	Safety Area
2A	Jan 1994	TLOF	**FATO**	Safety Area
2	Jan 1988	**FATO**	Takeoff and Landing Area	*No Definitions*
1B-C2	Jul 1986	Touchdown Pad	Takeoff and Landing Area	Peripheral Area
1B	Aug 1977	Touchdown Pad	Takeoff and Landing Area	Peripheral Area
1A	Nov 1969	Touchdown Area	Landing and Takeoff Area	Peripheral Area
1	Nov 1964	Touchdown Area	Landing and Takeoff Area	Peripheral Area
Original	Dec 1959	Touchdown Area	Takeoff and Landing Area	Peripheral Area



TLOF – Touchdown and Liftoff Area
FATO – Final Approach and Takeoff Area

U.S. Heliports Activated by Year (1959-Present)

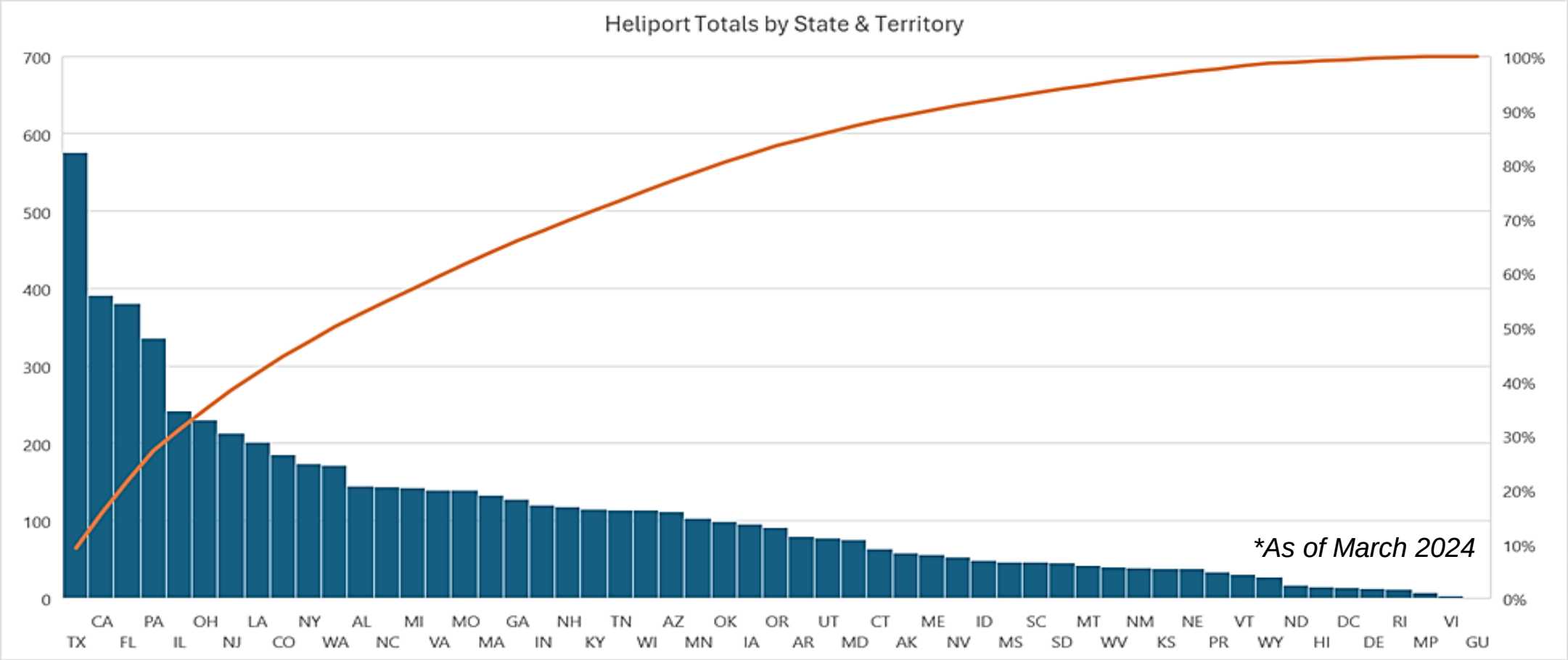


Classification Terms Over the Years

Classification	2D	2C	2B	2A	2	1B Chg-2	1B	1A	1	Org
GA	X	X	X							
Transport	X	X	X	X						
Hospital*	X	X	X	X						
PPR**	X	X	X							
Public					X	X	X			
Public GA				X						
Private				X	X	X	X			
Personal						X	X			
Private Class 1								X	X	
Public Class 2								X	X	
Public Class 3								X	X	
Heliport										X

**While the term "Hospital" has been in the FAA design guide since 1959, it wasn't until 2A that it was formally classified.*
***PPR stands for Prior Permission Required and is a military criteria not a classification. PPR is not defined in the CFR.*

Heliports by State & Territory

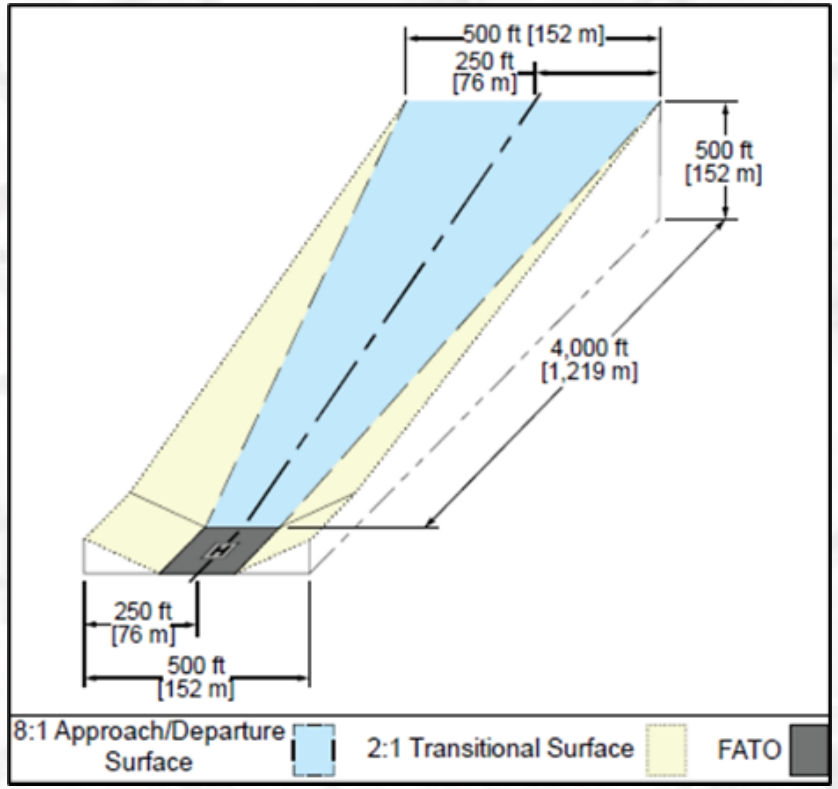


*Territories represented include Guam (GU), N Mariana Islands (MP), Puerto Rico (PR), and the Virgin Islands (VI).

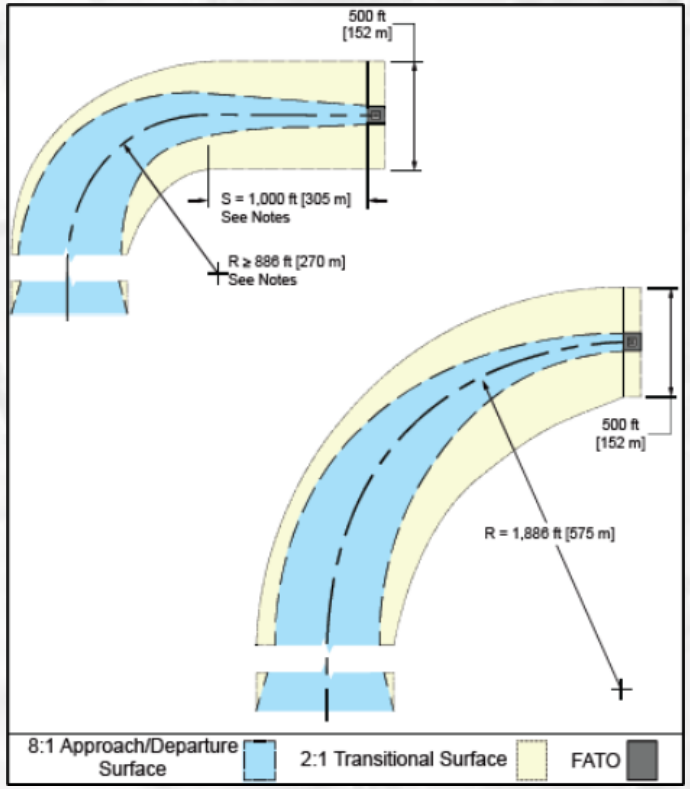
Heliports Airspace

SURFACE	DEFINITION
Primary	The area of the primary surface coincides in size and shape with the designated take-off and landing area. This surface is a horizontal plane at the elevation of the established heliport elevation.
Approach	The approach surface begins at each end of the heliport primary surface with the same width as the primary surface and extends outward and upward for a horizontal distance of 4,000 feet where its width is 500 feet. The slope of the approach surface is 8 to 1 for civil heliports and 10 to 1 for military heliports*.
Transition	These surfaces extend outward and upward from the lateral boundaries of the primary surface and from the approach surfaces at a slope of 2 to 1 for a distance of 250 feet measured horizontally from the centerline of the primary and approach surfaces.

Straight Approach



Curved Approach



Given the lack of airspace data capture in the AMR, Land-Use Compatibility Issues and Airspace Protection are much more difficult to accomplish.

**The 10-1 surface for military heliports listed does not align with current UFC 3-260-01 Standard*

Approach/Departure Paths

Loc Id	Runway Id	Length	Width
1M3	01/19	2700	100
14J	01/19	3050	75
GZH	01/19	5005	150
CQF	01/19	6604	100
M95	01/19	5009	80
4R3	01/19	5003	80
3M8	01/19	5144	80
GEU	01/19	7150	100
40G	01/19	4199	45
L25	01/19	2900	110

Airport App/Dep Headings

ADIP AMR DATA Comparison

Loc Id	Runway Id	Length	Width
MOB	H1	100	100
CHD	H1	100	100
FHU	H1	40	40
GYR	H1	64	64
OLS	H1	97	97
PAN	H1	50	50
SAD	H1	72	72
SEZ	H1	50	50
BIH	H1	40	40
CRQ	H1	40	50

Heliport App/Dep Headings

While FAA Form 7480 does have a data field to record a proposed heliport's approach/departure paths, FAA Form 5010 and the AMR do not have these same fields to capture this data. Additionally, the optional curved approach/departure path, which has been an option since 1959, is not captured in any data fields on either the 7480 or 5010 and is not provided for in the AMR.

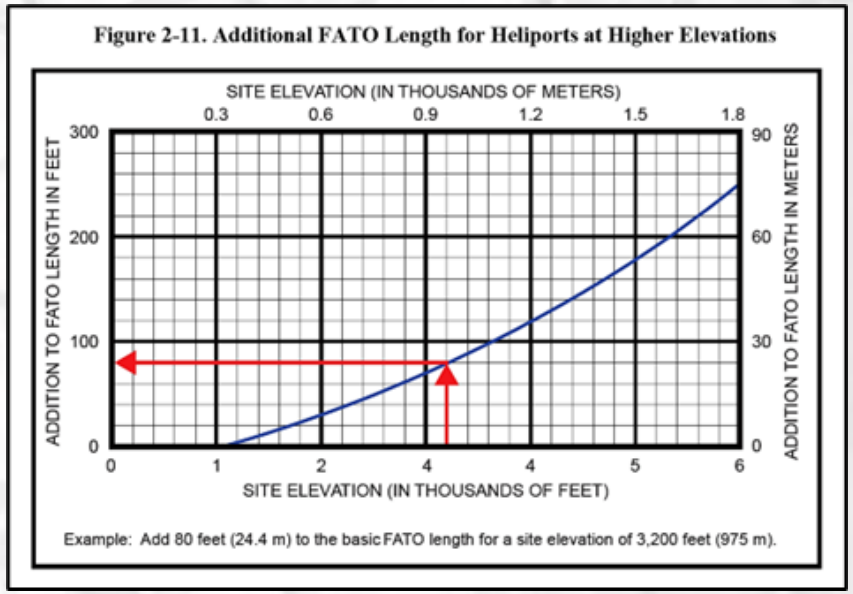
Note: A heliports TLOF, FATO and Safety Area are not identified, only Runway Length and Width. What this represents is not defined.

Heliport Performance Classification

At the current time there are no performance classifications for Heliports. The FAA conducted two specific studies for this purpose, however the recommendations were never adopted.

- F.D. Smith, DOT/FAA/RD-81/35, "Development of a Heliport Classification Method and an Analysis of Heliport Real Estate and Airspace Requirements", June 1981.
- Robert K. Anoll, Edwin D. McConkey, Robert J. Hawley, and Margaret B. Benton, DOT/FAA/RD-90/4, "Heliport VFR Airspace Design Based on Helicopter Performance", August 1991.

There is however a stipulation that for heliports above 1,000 feet in Elevation or greater it is recommended that their FATO size be increased. However, while this may provide a limited level of safety for certain types of aircraft at higher elevations, nowhere in the AMR or on FAA Forms 7480 or 5010 is this information captured or reflected.



Hover-Out-Of-Ground-Effect (HOGE)

In leu of increasing the size of the FATO for higher elevation heliports, or heliports on rooftops, the heliport design guide “2D” provides private use owners with the option to restrict the heliport to aircraft that are capable of Hover-Out-Of-Ground Effect (HOGE) performance for the conditions of the day.

FAA Reauthorization Act of 2024

SEC. 958. INFRASTRUCTURE SUPPORTING VERTICAL FLIGHT.

(a) UPDATE TO DESIGN STANDARDS.—The Administrator shall—

(2) not later than December 31, 2025, publish a performance-based vertiport design advisory circular; and

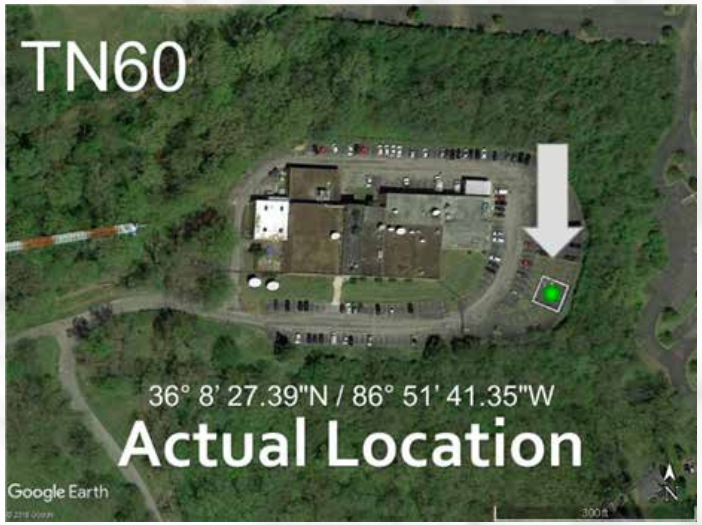
(3) begin the work necessary to update the advisory circular of the FAA titled "Heliport Design" (Advisory Circular 150/5390) in order to provide performance-based guidance for heliport design, including consideration of alternative fuel and propulsion mechanisms.

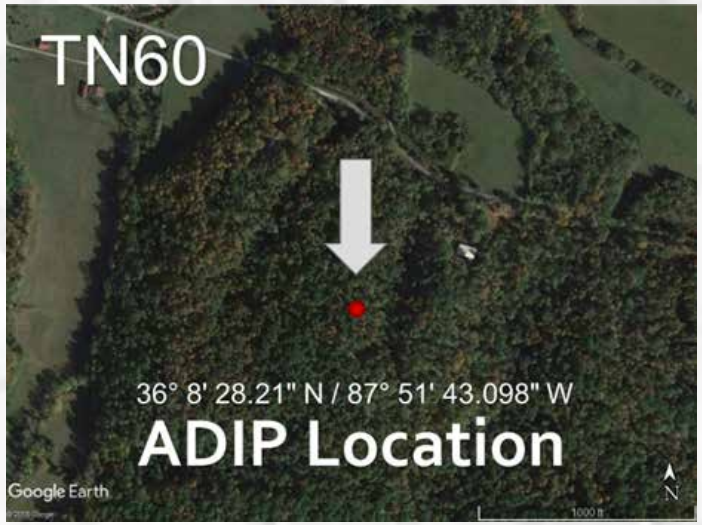
**At the current time no data fields exist to capture performance.*

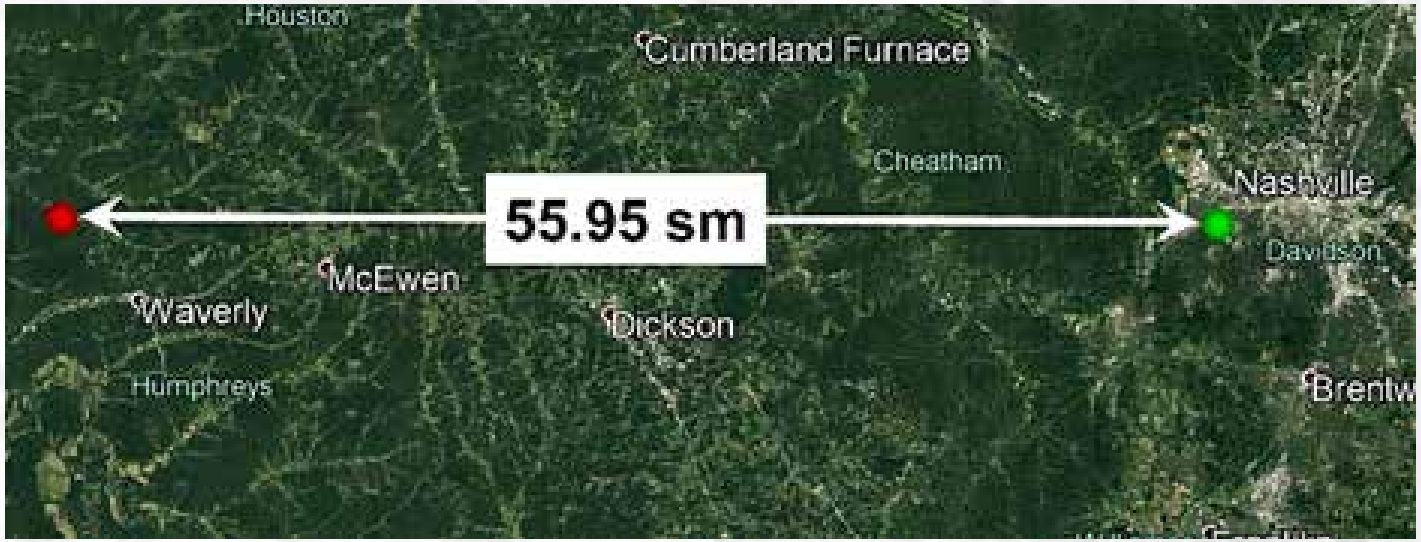
Location Accuracy

Example:

TN60 was activated in 1979, which would have been under heliport design guide “1B”. Since it is a private use facility the FAA has no legal oversight, authority, or responsibility for its information. Since it is only a recommendation that the data be updated and not a requirement it has existed with the incorrect coordinates for over 45 years.







Implications for UAS / BVLOS

Smart Mobility Corridor Project Ohio HWY- 33

*"This research will include both **air** and ground vehicles and will complement ongoing work to test **AUTONOMOUS** **and connected vehicles** along the 33 Smart Mobility Corridor, a 35-mile stretch of U.S. 33 between Dublin and East Liberty."*

Airspace Determinations

Title 14 CFR § 157.7.Determinations

Term	Definition
No Objection	(No Definition)
Conditional	A conditional determination will identify the objectionable aspects of a project or action and specify the conditions which must be met and sustained to preclude an objectionable determination.
Objectionable	An objectionable determination will specify the FAA's reasons for issuing such a determination.

FAA Dynamic Regulatory System (DRS) Determinations 8900.1

Term	Definition
No Objection	Use when there is no operational safety concern.
No Objection With Provision	Use when there is no operational safety concern (the proposal is generally acceptable to FS) but the inspector recommends specific changes or operational considerations.
Objection	Use when a proposal is clearly unacceptable due to operational safety concerns.

1. CFR Conditional, *"...conditions which must be met and sustained to preclude an objectionable determination"*. There are no provisions for tracking conditions for private use facilities.

1. Dynamic Regulatory System 8900.1 defines Objectionable as *"...clearly unacceptable due to operational safety"*. However, this unacceptability due to safety is not identified in the CFR definition.

Airspace Determinations

The FAA DRS evaluation standard state that:

"The designated Flight Standards Representative Heliport Evaluator determines if the owner/operator can conduct safe helicopter operations at the proposed location without undue risk to the operator, or to persons and property on the ground."

However, while the Flight Standards representative conducts the onsite evaluation, it is the respective Airport District Office (ADO) who writes the airspace determination letter. Situations have occurred where the Flight Standards representative has rendered an Objectionable determination, but the ADO has downgraded it to a Conditional determination with the justification that it is private use. No records were found that tracked or spoke to this process.

HAZARD Evaluations

FAA Procedures for Handling Airspace Matters, Order JO 7400.2P, states that a Heliport's 8:1 approach surface is identified as one of the surfaces to be considered in the evaluation and state that:

"A structure would have an **ADVERSE EFFECT** upon VFR air navigation if it penetrates the approach surface slope of any runway."

"ANY structure would have an **ADVERSE EFFECT** if it would exceed ANY of the heliport imaginary surfaces."

Example of Data Integrity Issues in Vertical Flight Infrastructure

Heliport/Helicopter Accident

Heliport:	AZ37
NTSB No.:	WPR22LA018
Date:	October 16, 2021
Location:	Tucson Medical Center Tucson, Arizona.

Accident Heliport Runway Information

In the NTSB accident investigation final report, under Airport Information, the NTSB lists Runway Length and Width of the Heliport as 100’ X 100’.

Airport Information

Airport:	TUCSON MEDICAL CENTER AZ37	Runway Surface Type:	Asphalt
Airport Elevation:	2485 ft msl	Runway Surface Condition:	Dry
Runway Used:	H1	IFR Approach:	None
Runway Length/Width:	100 ft / 100 ft	VFR Approach/Landing:	Full stop

Accident Heliport Runway Information

These same dimensions are reflected in the FAA AMR/ADIP system as well as the FAA ADIP Supplemental Heliport Data indicating that the TLOF at this heliport is 100’ X 100’ and ashplant and that there were NO Close-In Obstructions.

Helipad - H1 Ⓡ

Dimensions:	100 ft. x 100 ft.	Surface Type/Condition:	ASPH
Latitude:	32° 15' 5.87" N	Treatment:	
Longitude:	110° 52' 45.59" W	Edge Lights Intensity: Ⓡ	PERI

Ⓡ = Remark.

ARM ADIP Information

G	H	I	J	K
TLOF Length	**TLOF Width**	**Surface Type**	**Approach Lights**	**Close-In Obstruction**
100	100	ASPH	RLLS	N

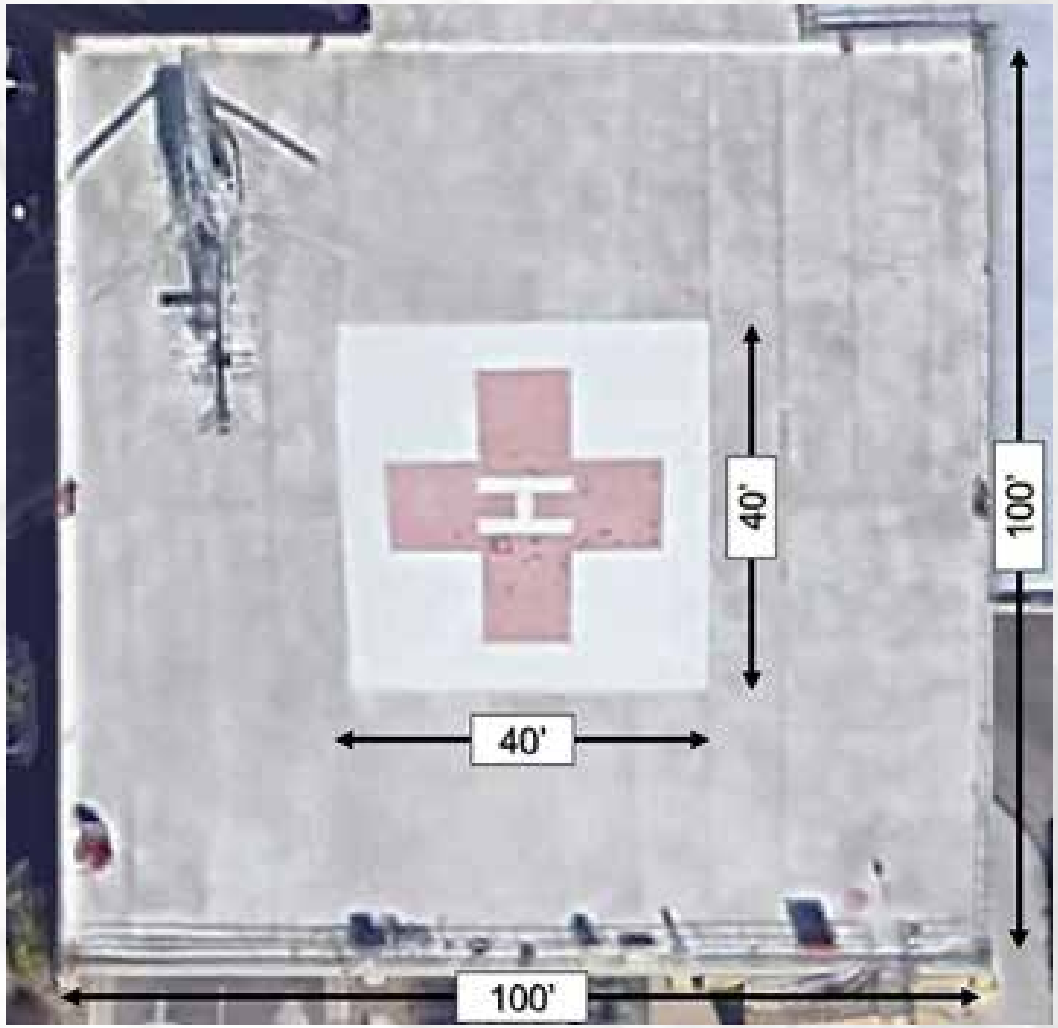
Supplemental Heliport Data

Reality

Upon closer examination of satellite imagery, the heliport was found to have a 40’ X 40’ TLOF which resides on a concrete parking garage deck measuring 100’ X 100’.

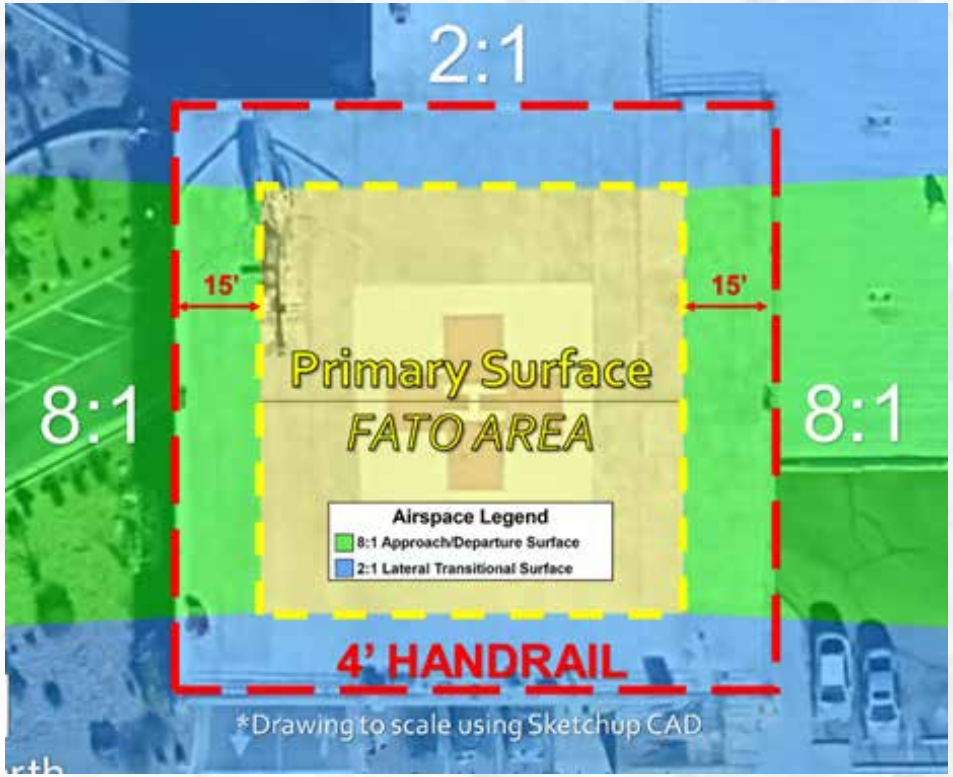
- Heliport activated Feb 1990.
- Design guide in effect 150/5390-2 (1988).

This was prior to the introduction of the term FATO, which was introduced in 1994, and the 1988 heliport guide did not define either the Peripheral Area or Safety Area.



Actual Heliport Surfaces

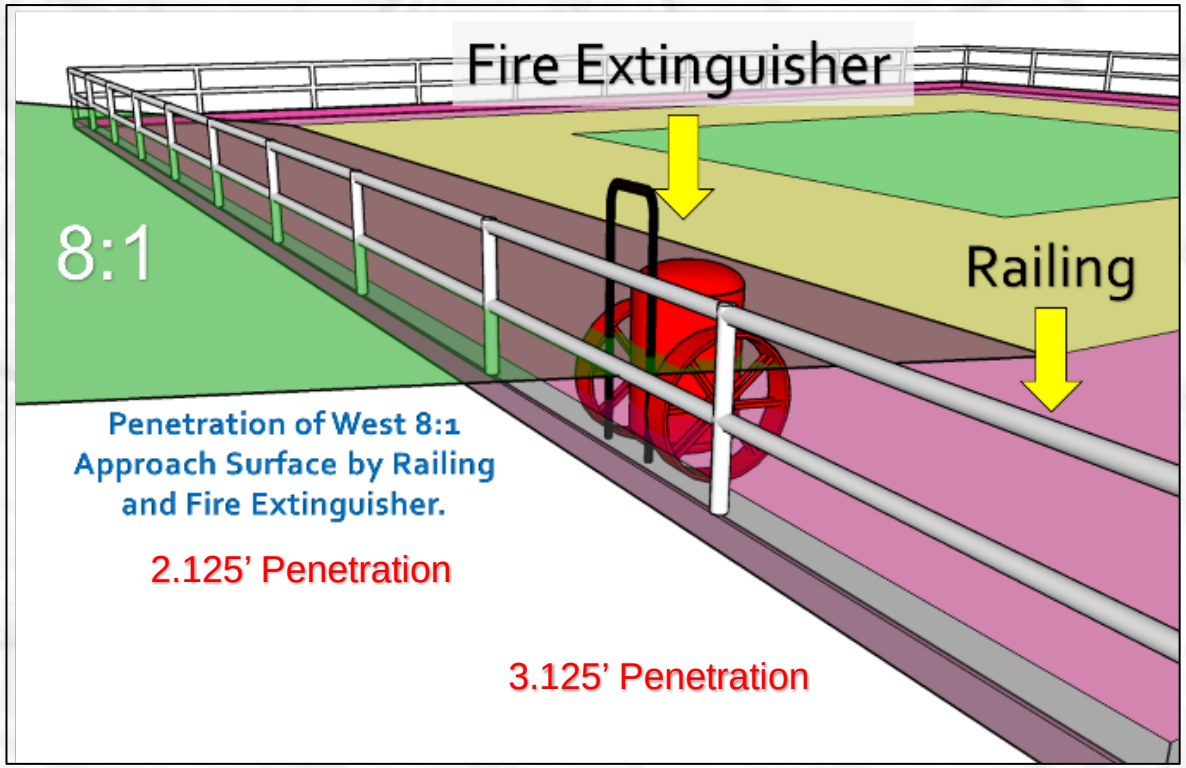
Dimensions Based on the accepted ratio
Overall Length = 1.2 X Rotor Diameter and
Rotor Diameter = 0.83 X Overall Length

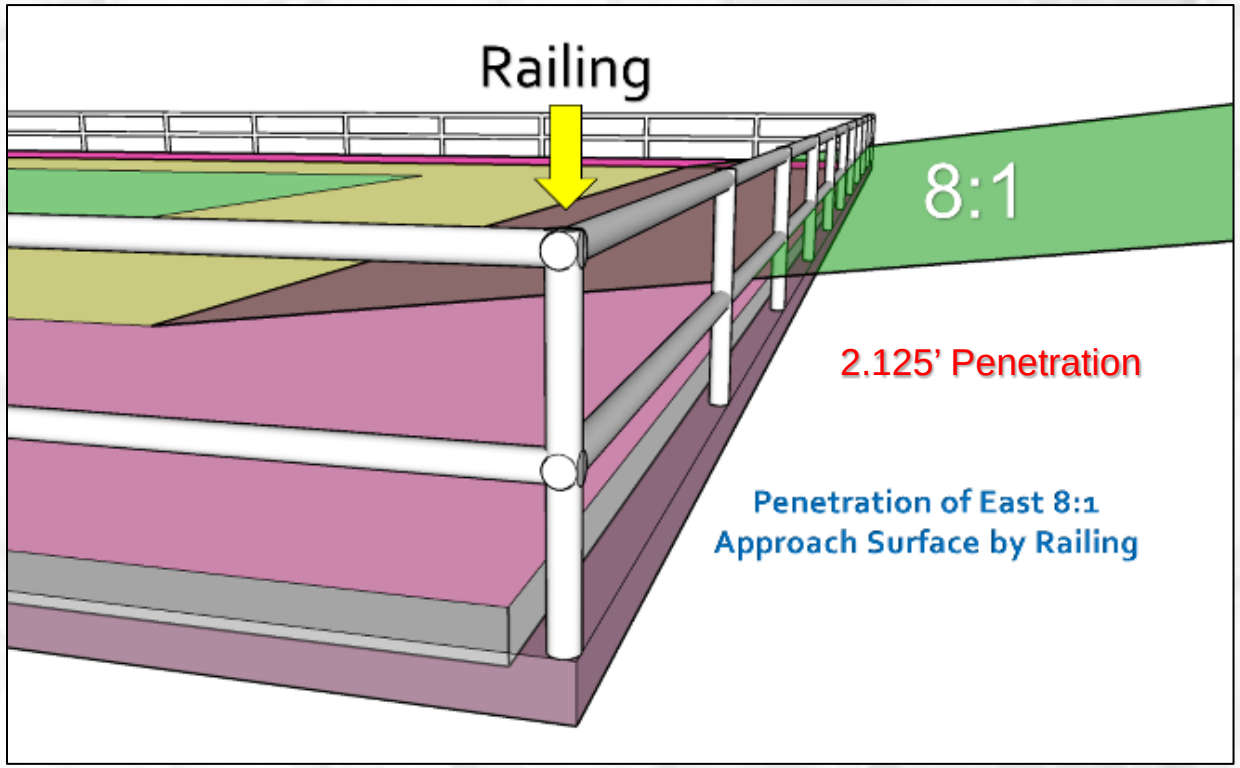


The AZ37 heliport incorporates a 4’ handrail around the perimeter of the 100’ X 100’ concrete surface, which penetrates the 8:1 approach surfaces.

ADVERSE EFFECT

ADVERSE EFFECT





For a 4' tall handrail or fence to be clear of a heliport or vertiport's 8:1 approach surface, it would need to be installed a minimum of 32' from the edge of the FATO. If the handrail or fence is located on terrain higher than the FATO, this distance would need to be increased. .

ADVERSE EFFECT

Accident Photos Curtesy of NTSB Docket Material

NTSB FINDINGS

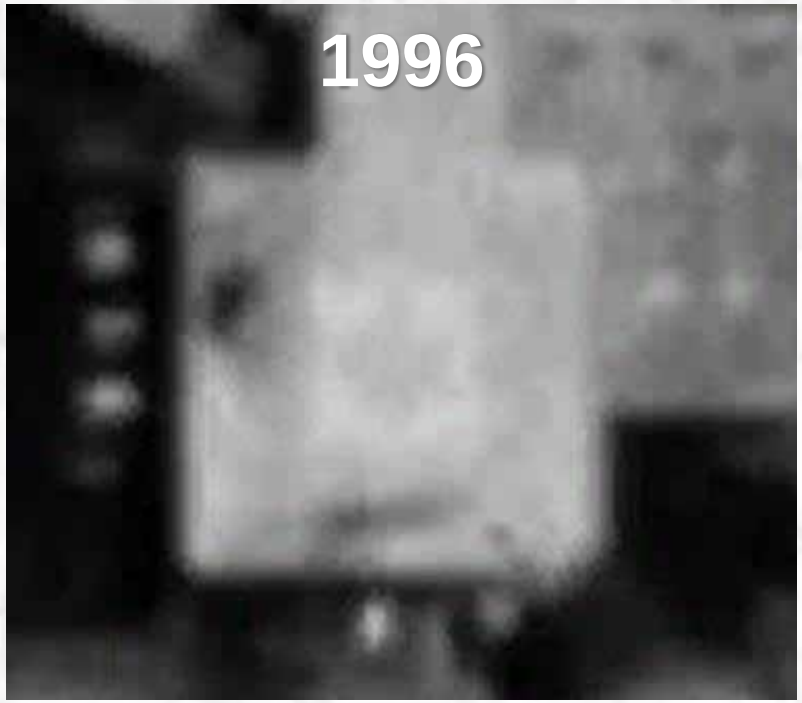
In their final report the National Transportation Safety Board (NTSB) determined the probable cause(s) of this accident to be:

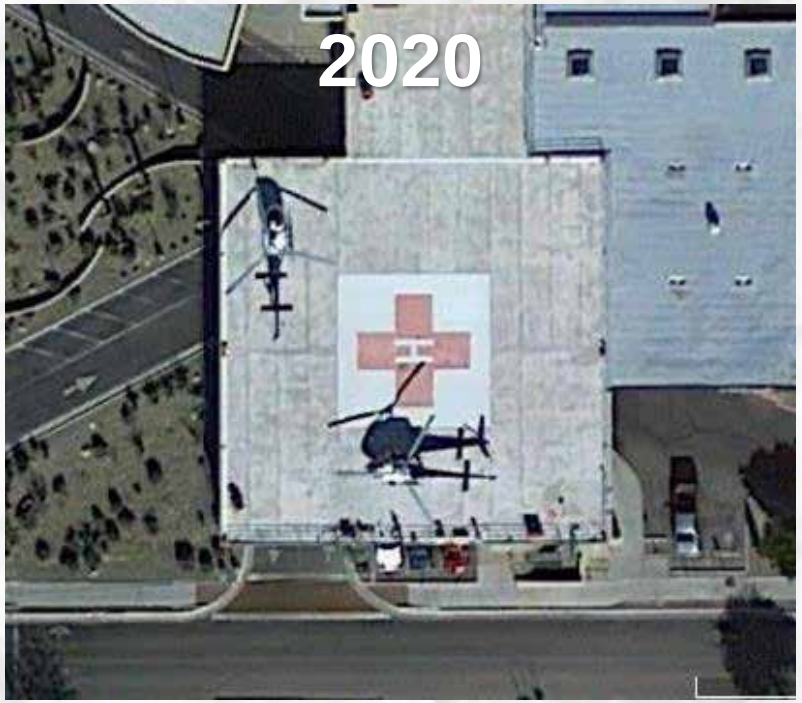
"The pilot's failure to maintain clearance from the ground equipment during landing."

There was no mention of the Fact that:

- The heliport did not meet published FAA standards.
- The information on file for the heliport in the FAA AMR/ADIP system was incorrect.
- The 8:1 approach surface was in fact penetrated by the handrail which in turn created an unsafe high-risk condition to exist causing an Adverse Effect.
- The fire extinguisher struck by the pilot on landing should have never been located on the heliport and also penetrate the 8:1 approach surface.
- It was a frequent practice for the certificate holder to have pilots land two and sometimes three helicopters within the 100' X 100' confines of this location on a routine bases since its activation.

CONTRIBUTING FACTOR





All FAA heliport design guides stipulate that a heliport is designed to support only one helicopter within the designated TLOF, FATO, and Safety Area.

CONCLUSIONS

1. A fundamental overhaul of how vertical flight infrastructure, both public-use and private-use, is evaluated, documented, and tracked in the United States is long overdue. High risk and unsafe conditions exist at vertical flight infrastructure today for which pilots are required to contend with on a daily basis and when an accident occurs the pilots are the ones held responsible rather than the infrastructure owners or aircraft certificate holders.
2. Accident investigators desperately need the appropriate training that provides them with the tools necessary to properly evaluate accidents that occur at vertical flight infrastructure so that infrastructure can be improved rather than allowed to continue to exist with unsafe and high-risk conditions.

CONCLUSIONS

3. The information contained within the FAA AMR for vertical flight infrastructure is in desperate need of validation and updating. While the ADIP systems that the FAA has developed provides an excellent platform for the collecting, updating, and data mining of information of infrastructure in the National Airspace System, the program needs additional staffing and funding to carry out the necessary effort to make the system effective and impactful.
4. Individuals and organizations looking to utilize existing heliports to support Advanced Air Mobility (AAM) operations must first verify that a heliport meets or is capable of meeting current standards. A change over to support different and unique aircraft with different performance capabilities will more than likely be considered a major change by the FAA.

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QUESTIONS

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THANK YOU